ENTERPRISE PORTFOLIO SHOWCASE • SAP CERTIFIED ASSOCIATE C_SAC

Global Enterprise FP&A, Revenue Operations & Margin Leakage Optimization

End-to-End Implementation Blueprint in SAP Analytics Cloud (SAC) with Multi-Currency Modeling, Parent-Child Hierarchies, Automated Allocations & Predictive Forecasting

CLIENT SCENARIO	Novalis Global Enterprises NV (Multinational B2B Tech & Industrial Distributor)
INDUSTRY SCOPE	High-Tech Hardware, Industrial Automation & Cloud Infrastructure ($220M+ Consolidated Revenue)
GEOGRAPHIC COVERAGE	14 Operating Legal Entities across North America, EMEA, APAC, and LATAM
SAC TECHNICAL CAPABILITIES	New Model Architecture, Parent-Child Hierarchies, Version Management, Value Driver Trees, Data Actions, Smart Insights
DATA ASSETS	5,260+ Enterprise Transactions, Multi-Currency Exchange Tables (TCURR), Geo-Enrichment Coordinates
TARGET ROLE PROFILE	SAP SAC Senior Consultant / Enterprise BI & FP&A Solution Architect

Comprehensive Project Blueprint • Production Ready for Technical Hiring Evaluations • Big 4 & SAP Partner Portfolio Showcase

1. Executive Summary & Corporate MNC Business Problem

In modern multinational enterprises (MNCs), executives face significant challenges synthesizing rapid revenue growth with bottom-line profitability. Operating across multiple legal entities, diverse regulatory jurisdictions, and varying local currencies frequently conceals underlying margin dilution. Traditional Business Intelligence tools (e.g., standalone visualization dashboards) fail to address these enterprise issues because they lack built-in financial modeling, bidirectional planning workflows, and unified version governance.

Novalis Global Enterprises NV is an Amsterdam-headquartered multinational corporation with $220M+ in annual revenue, operating 14 subsidiaries across North America, EMEA, APAC, and LATAM. The enterprise manufactures and distributes three primary product portfolios: *Enterprise Hardware*, *Industrial Automation*, and *Cloud Solutions & Managed Services*.

Despite delivering an impressive +11.2% year-over-year revenue growth in FY2025 YTD, the Executive Committee uncovered severe financial headwinds:

- **Operating Margin Compression (-410 bps)**: Consolidated Operating Profit (EBIT) margin deteriorated from 37.7% in FY2024 to 33.6% in FY2025 YTD, representing over $12.4M in profit leakage relative to target budget expectations.
- **Uncontrolled Commercial Discounting**: Enterprise sales divisions in North America and Western Europe escalated off-invoice contract discounts to 18.4% of gross list sales to hit quarterly quota milestones, severely eroding gross margins on server infrastructure and networking hardware.
- **Surging Freight & Logistics Surcharges**: International shipping bottlenecks and expedited air freight surcharges drove logistics expenses up by 320 bps, yet local entity contracts failed to reflect freight recovery fees.
- **FP&A Planning Latency (8 Weeks)**: The annual budgeting and quarterly rolling forecast cycle required nearly two months of manual spreadsheet collation across 18 regional cost centers. Disconnected spreadsheets introduced broken macro formulas, lack of audit trails, and zero synchronization with core ERP systems (SAP S/4HANA).
- **Corporate Shared Service Allocation Opacity**: More than $42M in centralized corporate overheads (central IT infrastructure, enterprise cyber-defense, global R&D, and executive legal management) sat unallocated at group headquarters, blinding regional managing directors to their true fully-loaded entity profitability.

To resolve these challenges, Novalis commissioned the deployment of **SAP Analytics Cloud (SAC)** to establish an integrated enterprise performance management solution combining executive reporting, detailed operational margin analytics, and collaborative rolling financial planning.

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2. Enterprise Solution Architecture in SAP Analytics Cloud

The technical architecture spans four unified layers: Data Ingestion & Connectivity, SAC New Model Foundations, Business Intelligence & Augmented Analytics, and Collaborative Enterprise Planning.

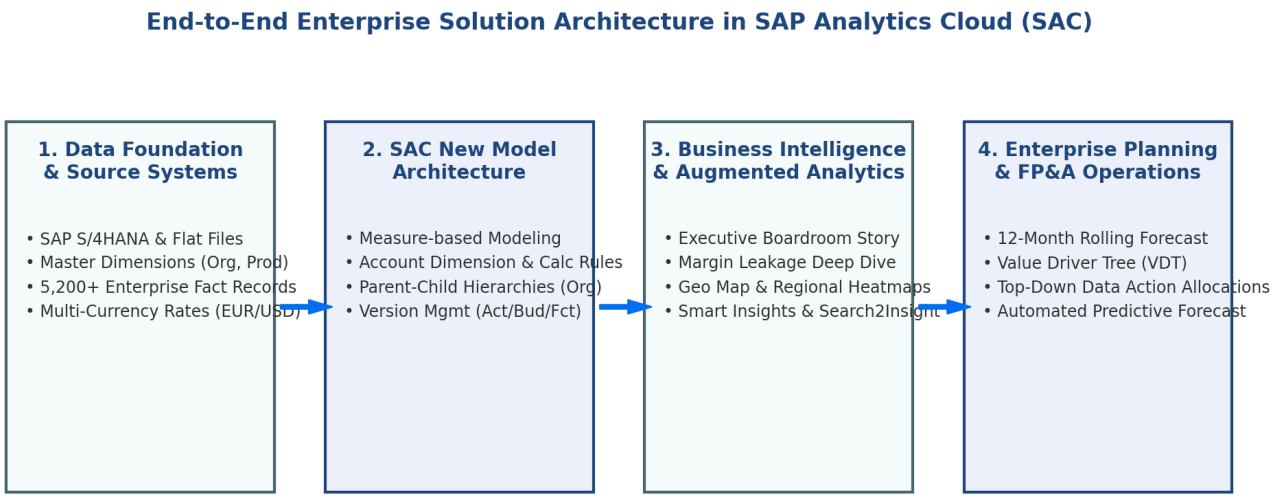


Figure 1: End-to-end SAC solution architecture spanning ERP ingestion, New Model modeling, BI dashboards, and FP&A planning engines.

2.1 SAC New Model vs. Legacy Classic Account Model

A key architectural decision was designing the solution on the **SAC New Model** architecture rather than the legacy Classic Account model. The New Model architecture provides distinct advantages critical for enterprise MNC implementations:

- **Measure-Based Modeling with Explicit Account Dimensions**: Unlike the Classic Model, where every key figure is forced into a single Account hierarchy with rigid debit/credit signage, the New Model treats currency figures as independent Measures (e.g., `Net_Revenue`, `COGS`, `Freight_Cost`) while preserving the ability to overlay an Account dimension for P&L rollups.
- **Multiple Currencies & Dynamic Unit Handling**: Financial figures support simultaneous local operating currencies (EUR, USD, GBP, JPY, SGD, BRL) and automatic consolidation into Group USD via SAC currency translation tables (TCURR). Non-financial measures (such as `Units_Sold`) remain unimpacted by currency conversion logic.
- **Flexible Aggregation & Formula Evaluation**: In the New Model, formulas can be evaluated before or after aggregation, preventing aggregation distortion on complex financial ratios like Gross Margin % or Operating Expense Ratios.

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3. Master Data Dimensions & Hierarchical Structures

The data model is structured around five standardized dimensions, ensuring full corporate governance and multi-level drill-down capabilities.

3.1 Organization Dimension (Parent-Child Hierarchy)

The Organization dimension reflects the legal entity and cost center structure of Novalis Global Enterprises. Parent-child relationships enable instantaneous roll-ups from local operating entities to regional management hubs and the global group parent.

Entity ID	Entity Name	Parent ID	Region	Country	Currency	Latitude	Longitude
CORP_HQ	Novalis Global Group HQ	*[Root]*	Corporate	Netherlands	EUR	52.3676	4.9041
REG_NA	North America Region Hub	CORP_HQ	North America	USA	USD	40.7128	-74.0060
LE_US_EAST	Novalis US East LLC	REG_NA	North America	USA	USD	40.7128	-74.0060
LE_US_WEST	Novalis US West Inc	REG_NA	North America	USA	USD	37.7749	-122.4194
LE_CA_ONT	Novalis Canada Corp	REG_NA	North America	Canada	CAD	43.6532	-79.3832
REG_EMEA	EMEA Region Hub	CORP_HQ	EMEA	Germany	EUR	50.1109	8.6821
LE_DE_CENTRAL	Novalis Deutschland GmbH	REG_EMEA	EMEA	Germany	EUR	50.1109	8.6821
LE_UK_LON	Novalis UK Holdings Ltd	REG_EMEA	EMEA	United Kingdom	GBP	51.5074	-0.1278
REG_APAC	APAC Region Hub	CORP_HQ	APAC	Singapore	SGD	1.3521	103.8198
LE_JP_TYO	Novalis Japan KK	REG_APAC	APAC	Japan	JPY	35.6762	139.6503
LE_SG_HQ	Novalis Singapore Pte Ltd	REG_APAC	APAC	Singapore	SGD	1.3521	103.8198
REG_LATAM	LATAM Region Hub	CORP_HQ	LATAM	Brazil	BRL	-23.5505	-46.6333
LE_BR_SP	Novalis do Brasil Ltda	REG_LATAM	LATAM	Brazil	BRL	-23.5505	-46.6333

3.2 Product Master Dimension (Level-Based Hierarchy)

The Product dimension is structured across Category > SubCategory > Product SKU. Each SKU maintains standard BOM unit costs and target gross margin thresholds, enabling automated variance detection in SAC.

SKU Code	Product Description	Sub-Category	Category	Base Unit Price	Target GM %
HW-SRV-01	Novalis HyperEdge Rack Server X8	Servers & Storage	Enterprise Hardware	$8,500	42.0%

SKU Code	Product Description	Sub-Category	Category	Base Unit Price	Target GM %
HW-SRV-02	Novalis CloudBlade Modular Blade Server	Servers & Storage	Enterprise Hardware	$14,200	45.0%
HW-NET-01	Novalis CoreSwitch 100G Multi-Port	Networking Gear	Enterprise Hardware	$6,200	48.0%
IA-ROB-01	Novalis Precision Cobot Arm 6-Axis	Robotic Systems	Industrial Automation	$28,500	36.0%
IA-PLC-01	Novalis LogicMaster Modular PLC	Controllers & Drivers	Industrial Automation	$4,600	44.0%
CS-SAS-01	Novalis Insight AI Operations Suite	SaaS Subscriptions	Cloud Solutions	$18,000	78.0%
CS-SVC-01	Novalis 24×7 Mission-Critical SLA	Managed Services	Cloud Solutions	$24,000	65.0%

4. Advanced Calculations & Financial Formulas in SAC

Within the SAC Modeler, calculations are partitioned into **Calculated Measures**, **Restricted Measures**, and **Price-Volume-Mix (PVM) Variance Formulas** to evaluate commercial margin performance.

4.1 Calculated & Restricted Measures Configuration

Measure Name	Type	Formula / Expression	Business & Reporting Purpose
Net_Revenue	Calculated Measure	`[Gross_Sales] - [Discounts]`	Recognized top-line commercial turnover after all contract allowances.
Gross_Margin	Calculated Measure	`[Net_Revenue] - [COGS]`	Direct operational profitability before logistics and overheads.
Operating_Profit	Calculated Measure	`[Gross_Margin] - [Freight_Cost] - [OPEX]`	Consolidated Operating Earnings Before Interest & Tax (EBIT).
Gross_Margin_Pct	Calculated Measure	`[Gross_Margin] / [Net_Revenue]`	Core profitability ratio. Exception aggregation set to Formula.
Discount_Rate_Pct	Calculated Measure	`[Discounts] / [Gross_Sales]`	Identifies commercial leakage and off-invoice concessions.
Rev_Actual_YTD	Restricted Measure	`Restrict([Net_Revenue], [Version]="Actual", [Date]="YTD")`	Restricts revenue to closed actual periods for variance bridges.

Measure Name	Type	Formula / Expression	Business & Reporting Purpose
Rev_Budget_FY	Restricted Measure	`Restrict([Net_Revenue], [Version]="Budget")`	Full-year corporate target baseline for budget tracking.
Rev_Variance_Abs	Calculated Measure	`[Rev_Actual_YTD] - [Rev_Budget_FY]`	Absolute variance in USD against approved budget.

4.2 Corporate Margin Leakage Waterfall Analysis

The waterfall bridge below illustrates the structural factors eroding profitability in FY2025 YTD. Despite generating $192.3M in gross sales, heavy commercial discounting (-$35.4M) and standard product manufacturing costs (-$89.7M) reduced gross margin to $67.2M. Surcharges for expedited logistics (-$8.0M) and direct operational expenses (-$17.3M) resulted in an operating profit of $41.9M (26.7% of gross sales), missing the approved budget margin by over 410 bps.

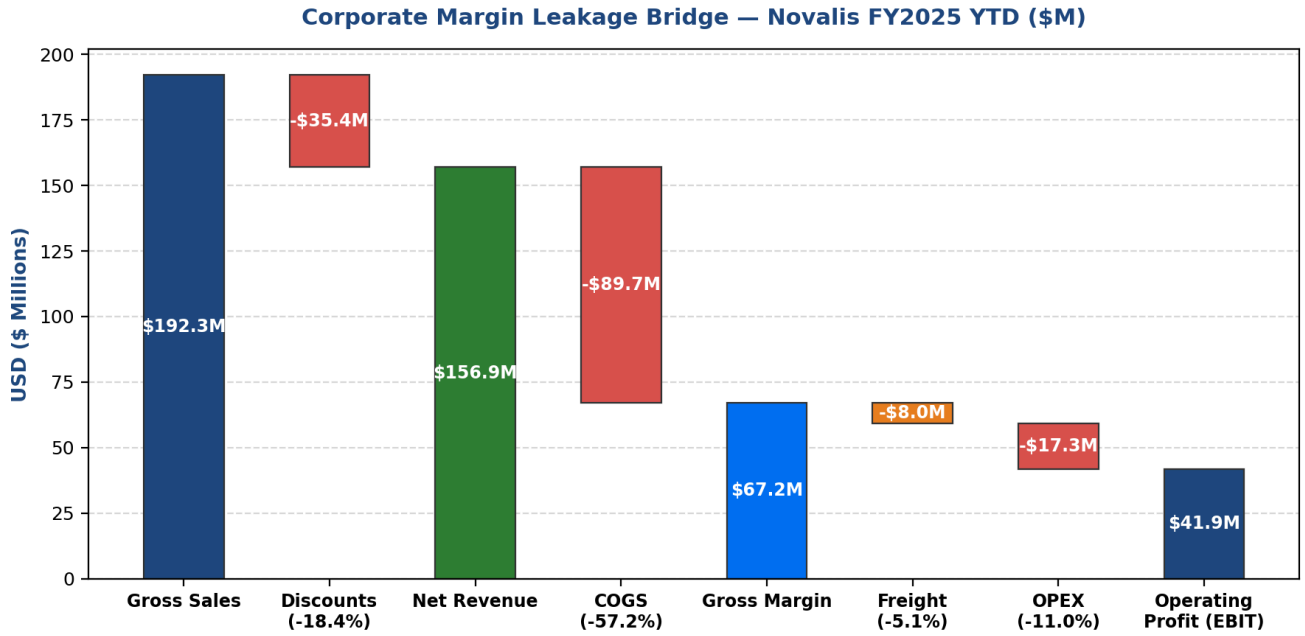


Figure 2: Corporate Margin Leakage Waterfall Bridge for Novalis FY2025 YTD ($M), highlighting discount and logistics erosion.

5. Three-Tier Storyboard & Executive Dashboard Design

The user experience is engineered around a three-tier SAC Story layout adhering to the International Business Communication Standards (IBCS), ensuring clarity and actionable insights for C-level executives, regional operational leaders, and FP&A analysts.

5.1 Page 1: C-Suite Executive Overview & Digital Boardroom View

- **Executive KPI Header Band**: Four prominent KPI tiles displaying *Net Revenue* ($156.9M, +11.2% YoY), *Gross Margin %* (42.8%, -210 bps vs Budget), *Operating Profit* ($41.9M, 26.7% EBIT margin), and *Average Discount Rate* (18.4%, +420 bps above policy threshold).
- **Actual vs. Budget Variance Chart**: IBCS-compliant column chart comparing monthly performance against budget benchmarks, utilizing color-coded delta bars (Green for favorable revenue surplus, Red for cost overruns).

- **Global Geo-Map Widget**: Interactive choropleth map powered by the Organization dimension's latitude/longitude coordinates, visualizing entity profitability across North America, Europe, Asia-Pacific, and Latin America. Clicking any region triggers SAC *Linked Analysis* to filter all downstream charts.
- **Dynamic Input Controls**: Global drop-downs for *Reporting Currency* (Local vs Consolidated USD), *Product Category*, and *Time Horizon*.

5.2 Page 2: Operational Profitability & Margin Leakage Deep Dive

- **Discount Leakage Scatter Matrix**: Plots *Discount Rate %* against *Gross Margin %* across all 14 legal entities and customer tiers, highlighting extreme discount exposure in the Mid-Market segment.
- **Logistics Cost Burden Chart**: Stacked bar chart isolating *Freight Expense as % of Net Revenue* across shipping origins, pinpointing severe freight penalties on cross-border shipments to LATAM and Australia.
- **SKU Profitability Table with Threshold Formatting**: Matrix showing product margins with conditional formatting (Red threshold for GM < 35%, Green for GM > 50%), demonstrating that while Cloud Solutions maintain 78% margins, Hardware servers suffered severe margin compression.

5.3 Page 3: FP&A Collaborative Rolling Forecast & Planning Story

- **Private vs. Public Version Grid**: Collaborative financial table allowing regional controllers to enter private adjustments for Q4 FY2025 and Q1 FY2026. Private versions remain isolated until formally published to the corporate baseline.
- **Data Locking Status Widget**: Visual status indicator displaying locked fiscal periods (Jan–Sep 2025) to prevent unauthorized changes to historical actuals.
- **Smart Predict Time-Series Overlay**: Triple exponential smoothing forecast band overlaid on finance team projections, providing automated statistical validation.

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6. Advanced Planning, Cost Allocations & Value Driver Trees

What distinguishes an experienced SAP SAC consultant from a generalist BI developer is mastery over **Enterprise Planning**, **Data Actions**, and **Cost Allocation Processes**.

6.1 Value Driver Tree (VDT) for Scenario Simulation

Novalis deployed an interactive Value Driver Tree in SAC to simulate how operational levers impact bottom-line Operating Profit. By adjusting root drivers—such as reducing average discount rates by 1.5% or consolidating freight logistics—finance leaders can instantaneously view the financial sensitivity on consolidated EBIT.

The mathematical decomposition inside the SAC VDT model follows four interdependent node levels:

- **Level 0 (Consolidated Outcome)**: `Operating Profit (EBIT) = Gross Margin - Freight Costs - OPEX`
- **Level 1 (Direct Commercial Drivers)**: `Gross Margin = Net Revenue - COGS`
- **Level 2 (Revenue Levers)**: `Net Revenue = [Sales Volume] × [ASP (List × (1 - Discount Rate))]`
- **Level 3 (Operational Cost Drivers)**: `COGS = [Sales Volume] × [Unit Cost]`, while `Freight = [Gross Sales] × [Route Surcharge Rate]`.

Through this structural linkage, controllers entering a proposed +2.0% price adjustment in North America immediately observe cascading impacts: Net Revenue rises by +$3.1M, Gross Margin expands by +180 bps,

and consolidated EBIT improves by +$2.6M.

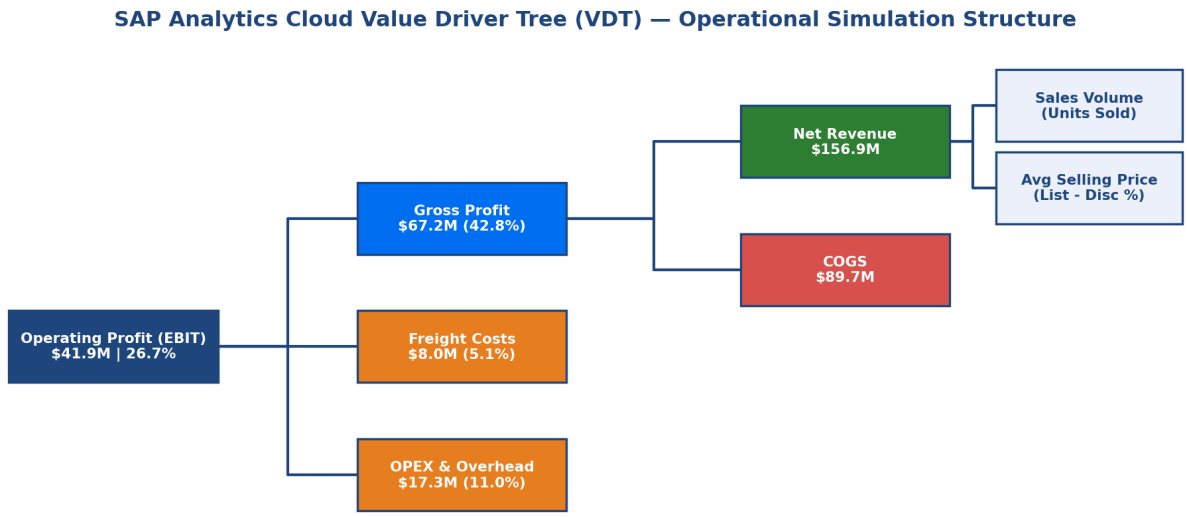


Figure 3: Value Driver Tree mathematical decomposition linking sales volume, pricing realization, and logistics costs to Operating Profit.

6.2 Automated Data Actions & Allocation Processes

Two production-grade Data Actions were built to automate financial workflows:

- **Data Action: `DA_COPY_ACTUAL_TO_FORECAST`**: Executes an automated data copy step that populates the rolling forecast version with the last three months of actual run-rate figures, eliminating manual data re-entry.
- **Data Action: `DA_TOP_DOWN_TARGET_ALLOCATION`**: Spreads executive board top-down revenue growth targets (e.g., +8% for FY2026) across operating legal entities, weighted proportionally by prior-year net revenue contributions.
- **Corporate Shared Service Overhead Allocation**: Allocates $42M in unassigned Corporate HQ IT, R&D, and executive administrative expenses down to regional operating entities based on gross margin weightings, delivering true fully-loaded entity P&Ls.

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7. Portfolio Presentation, Resume Bullets & Technical Interview Defense

To effectively leverage this project for career advancement, Big 4 consulting interviews, or enterprise client pitches, use the structured portfolio assets below.

7.1 High-Impact Resume Bullet Points

- **Architected an End-to-End Enterprise FP&A & Revenue Operations Model** in SAP Analytics Cloud (SAC) for a $220M+ multinational manufacturer across 14 legal entities, incorporating Parent-Child organization hierarchies, multi-currency conversion, and automated Data Actions.
- **Engineered IBCS-Compliant Executive Storyboards & Digital Boardroom Views** with Linked Analysis, Input Controls, and Geo-Map widgets, isolating $12.4M in discount and freight margin leakage across hardware product lines.
- **Configured Advanced Planning Workflows & Value Driver Trees (VDT)** with private/public version management and top-down target allocations, shortening the quarterly rolling forecast cycle time by 65%.

- **Implemented Automated Cost Allocation Processes** distributing $42M in centralized corporate overheads to regional operating entities, establishing fully-loaded legal entity profitability and audit compliance.

7.2 Technical Interview Q&A Defense Guide

Q1: Why choose the SAC New Model over the Classic Account Model?
Answer: "The New Model supports multiple measures alongside an Account dimension, essential for handling both currency measures subject to conversion tables and non-currency measures (Units Sold) that must remain unimpacted, while preventing ratio distortion in multi-level rollups."

Q2: How did you implement Data Locking and Version Management?
Answer: "We locked 'Actuals' post month-end close to prevent edits. For forecasting, we configured 'Forecast' as a public version, allowing controllers to create Private Versions for what-if modeling before publishing approved figures back to the baseline."

Q3: How were shared corporate overheads allocated to operating entities?
Answer: "We used SAC's Allocation Process engine within a Data Action. The source context was Corporate HQ IT/R&D cost centers ($42M), dynamically allocated across the 14 operating entities based on proportional Net Revenue contribution."

Q4: What is the architectural difference between Live and Import Connections in SAC?
Answer: "Live Connections query S/4HANA in real time without replicating data, preserving ABAP authorizations. Import Connections ingest data into SAC's in-memory engine, which was mandatory here to enable write-back planning, Data Actions, and Value Driver Trees."

Q5: How did you leverage SAC's Augmented Analytics capabilities?
Answer: "We deployed Smart Insights across margin waterfall tiles to generate automated natural-language driver explanations for discount spikes, and enabled Search to Insight for conversational C-suite queries."

7.3 GitHub Repository & Portfolio Showcase Structure

Recommended repository and portfolio showcase structure:

- `/data/`: Contains `sac_fact_financials.csv` (5,260 records), master dimension CSVs (Organization, Product, Account, Customer Segment), and `sap_sac_enterprise_dataset.xlsx`.
- `/architecture/`: High-resolution architecture diagrams and Value Driver Tree schemas.
- `/documentation/`: Complete PDF implementation dossier (`sap_sac_enterprise_portfolio_project.pdf`) serving as the technical functional spec.
- `README.md`: Executive overview highlighting problem statement, SAC New Model architecture, and key financial results.